

DS 207: Mathematics for Data Science

General Course Guidelines

1. Grading System

1.1 Exercises — 25%

- Exercises in the lesson (Jupyter) notebooks
- To be given to students at least a week prior to the class for that topic
- Jupyter notebook with answers to be submitted/saved in the student's respective shared Google Drive folder
- File name: ds207_<exercise_number>_<topic_title>_<last_name> as suffix to the notebook's original file name (e.g., `ds207_01_descriptivestatistics_corro.ipynb`)
- Deadline of submission: 11:59 PM of the day after the session when the discussion of the associated topic was finished, OR any date specified by the lecturer

1.2 Mini-Project — 25%

- Phase 1: Data + Proposal including objective, numerical analysis connection, business relevance, preliminary dataset
- Phase 2: Results + analysis including experiments, error analysis, interpretation, reflection
- Detailed instruction will be given in one of the future sessions
- Weeks 14–15 are reserved as unstructured work time for Mini-Project Phases 1–2 (see Course Outline)

1.3 Class Participation — 25%

- 5-point scale averaged
 - 2 points - attempted but incorrect
 - 3 points - correct answer
 - 4 or 5 points - correct answer + relevant insights (depends on the quality of the answer and insight)
 - Valuable insight may refer to providing an accurate answer to the question and connecting it to actual personal experience at work or real-world use cases
- Finding crucial/significant conceptual errors in the lesson notebook is considered for the class participation marks. A full mark is given provided an explanation as to why it is a conceptual error is given.
- Recommendation: Target at least 2 meaningful contributions per session

1.4 Major Exams — 25%

- There will be 3 major exams (Exam 1, Exam 2, Exam 3/Final), each weighted equally
- Each exam contributes an equal one-third of the 25% Major Exams component (~8.33% of the total course grade each)
- Each exam is a 3-hour session; approximately 20–25% of each exam is conceptual/reasoning-based, requiring no calculation

2. Course Outline

Wk	Dates	Topic / Activity
1	Aug 24–28	Descriptive statistics, data types
2	Aug 31–Sep 4	Distribution shape & skewness + Missing data, outliers, transformations
3	Sep 7–11	Correlation analysis & multicollinearity + EDA synthesis; review
4	Sep 14–18	EXAM 1 (3-hr)

Wk	Dates	Topic / Activity
5	Sep 21–25	Probability foundations, Bayes' theorem
6	Sep 28–Oct 2	Random variables/distributions + Sampling distributions/CLT
7	Oct 5–9	Estimation, confidence intervals
8	Oct 12–16	Hypothesis testing framework; review
9	Oct 19–23	EXAM 2
10	Oct 26–30	Resampling foundations, bootstrap + Permutation tests
11	Nov 2–6	Linear algebra I — vectors, matrix operations, systems
12	Nov 9–13	Linear algebra II — eigenvalues, eigenvectors
13	Nov 16–20	Synthesis — correlation matrices meet eigendecomposition; review
14	Nov 23–27	Buffer – for term project
15	Nov 30–Dec 4	Buffer – for term project
16	Dec 7–10	EXAM 3 (FINAL) - cumulative

3. Use of Generative AI Tools

- Allowed: concept clarification, pseudocode, debugging, idea generation.
- Crucial Responsibility:
 - Always verify accuracy of generated answer
 - Understand the idea and concept behind the generated contents. It may come in handy during recitations and examinations when the use of AI is not allowed.

4. Citation and Attribution

To maintain academic integrity and professional standards in data science, all students must properly cite references, datasets, and the use of Generative AI tools in all assignments, reports, notebooks, and the mini-project.

Students must cite all external sources used, including:

- Textbooks
- Academic papers
- Online articles or documentation
- Lecture notes from other classes
- Code snippets taken or adapted from external sources

5. University Policies and Guidelines

All relevant University-wide academic and administrative guidelines apply to this course.